Tiny drones that can operate in cramped spaces


New generation of tiny, agile drones (Credit: https://news.mit.edu)

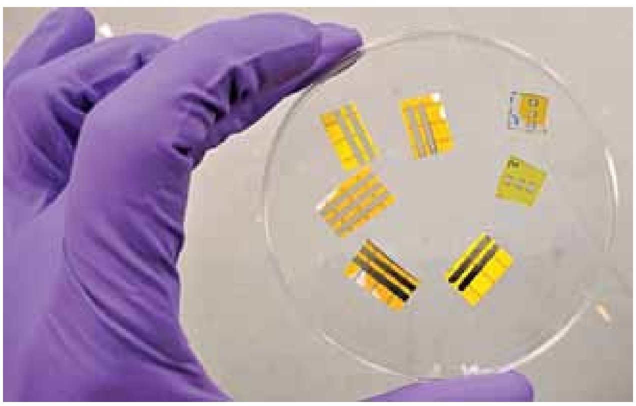
Insects are remarkably acrobatic and resilient in flight, which allows them to navigate through wind gusts, obstacles, and general uncertainties. Such traits are hard to incorporate into flying robots. However, at the Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, an insect-sized drone has been developed that is quite dexterous and resilient. The aerial robot is powered by a soft actuator made of thin rubber cylinders coated in carbon nanotubes. When voltage is applied to them, an electrostatic force is produced that squeezes and elongates the rubber cylinder. Repeated elongation and contraction cause the drone's wings to beat fast. It is hoped that the tiny robots could one day aid humans in industry and agriculture.

Small gadgets powered by the human body

Nanoengineers at the University of California San Diego have developed a wearable 'microgrid' that harvests and stores energy from the human body to power small electronics. The wearable is built from a combination of flexible electronic parts and consists of sweat-powered biofuel cells, motion-powered triboelectric generators, and energy-storing supercapacitors. All parts are flexible, washable, and can be screen printed onto clothing. Just like a city microgrid integrates a variety of local, renewable power sources, a wearable microgrid integrates devices that locally harvest energy from different parts of the body, like sweat and movement, while containing energy storage.


The wearable microgrid that uses energy from human sweat and movement to power an LCD wristwatch and electrochromic device (Credit: https://ucsdnews.ucsd.edu)

Smart OLED tattoo for TV and smartphone screens


Light-emitting smart tattoo (Credit: www.ucl.ac.uk)

Scientists at the University College London (UCL) and the Istituto Italiano di Tecnologia (Italian Institute of Technology) have created a temporary tattoo with light-emitting technology that can be used in TV and smartphone screens, paving the way for a new type of 'smart tattoo' with a range of potential uses. The technology uses organic light-emitting diodes (OLEDs) and is applied in the same way as water transfer tattoos. That is, the OLEDs are fabricated on to temporary tattoo paper and transferred to a new surface by pressing on to it and dabbed with water. The process could be combined with other tattoo electronics to emit light in case of dehydration or to get out of the sun to avoid sunburn. OLEDs could be tattooed on packaging or fruit to signal when a product has passed its expiry date, or will soon become inedible. These could also be used for fashion in the form of glowing tattoos.